

Enhancing Clinical Reasoning in CUREBench Using a Structured Tool-Augmented Pipeline

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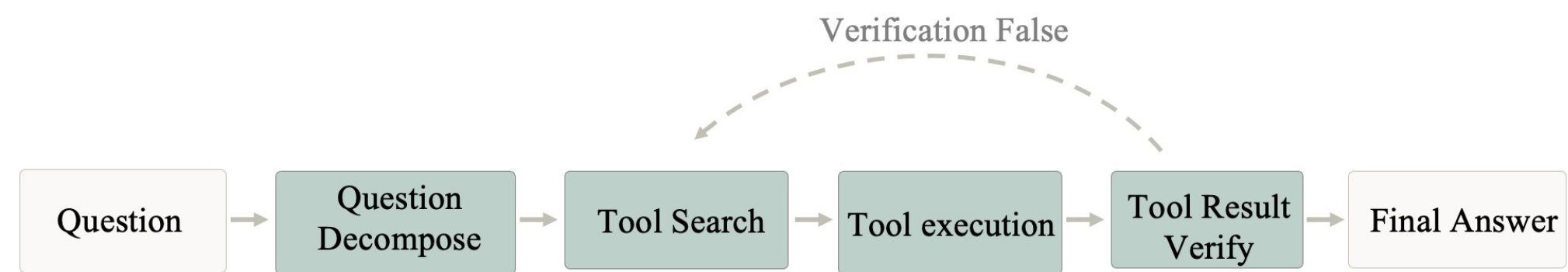
1. Introduction

CUREBench is a benchmark designed to assess AI models on clinical reasoning and therapeutic decision-making through diverse, clinically oriented tasks such as diagnosis support, treatment planning, and drug safety assessment.

In this study, we apply GPT-5, a state-of-the-art API-based large language model, to the CUREBench tasks to explore its applicability in biomedical and clinical domains. Our approach examines how general-purpose LLMs can be adapted to specialized medical tasks and shows how incorporating medical tools can enhance the model’s medical reasoning performance.

2. Approach

We enhanced the clinical reasoning performance of the LLM by applying a Structured Reasoning Pipeline and integrating tool-based methods.



Structured Reasoning Pipeline

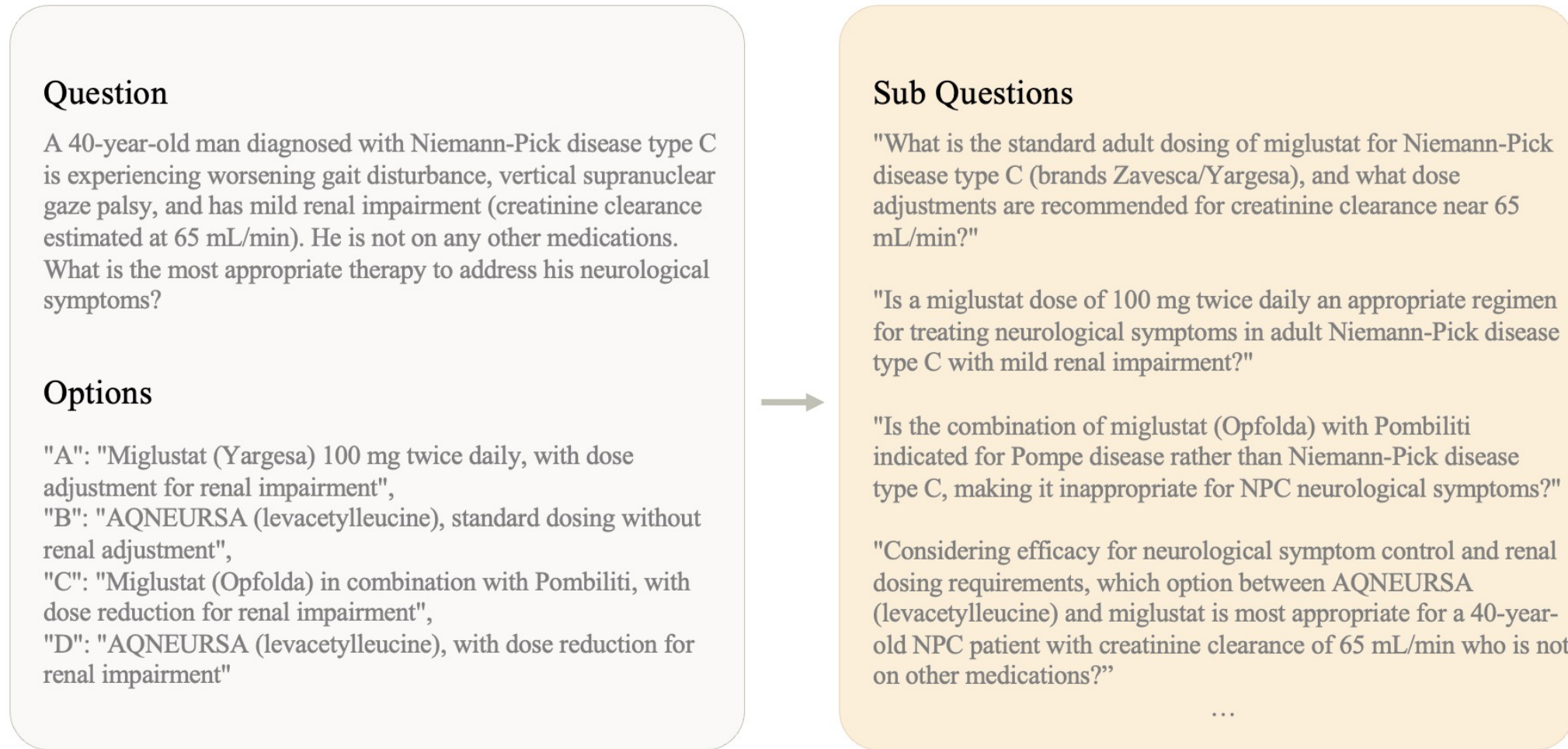
- Decompose the original question into smaller sub-questions.Bulleted list item.
- Use tools to retrieve or generate the necessary information for each sub-question.Bulleted list item.
- Integrate the collected information to improve the accuracy of the final answer.

Tool Utilization

- Use each sub-question as a query and perform tool search via ToolRAG over the ToolUniverse.
- Execute the selected tool and evaluate the result through a verification step.
- Repeat the search–execution–verification cycle until a useful and reliable tool result is obtained.

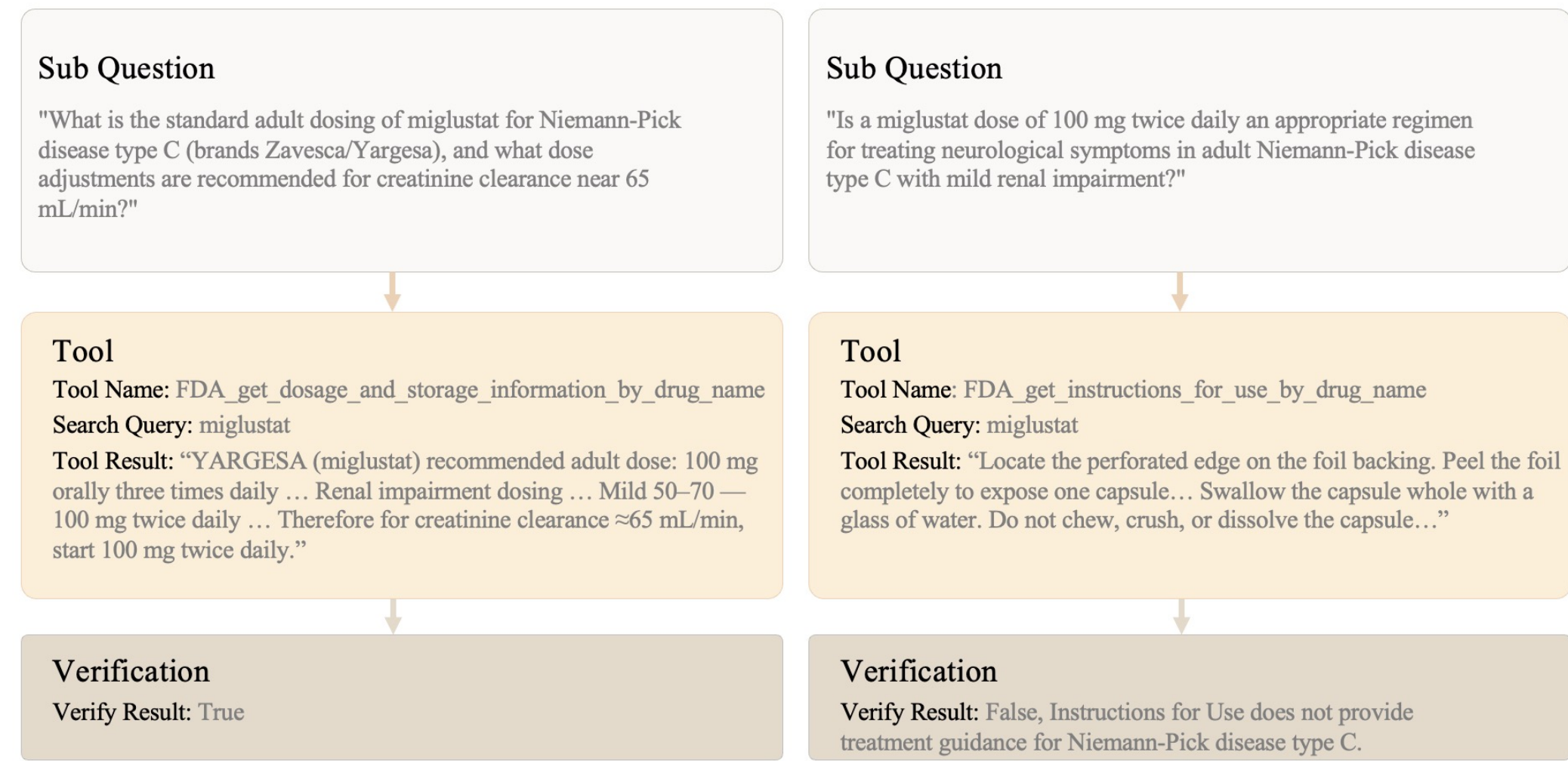
3. Method Overview

Decompose Question



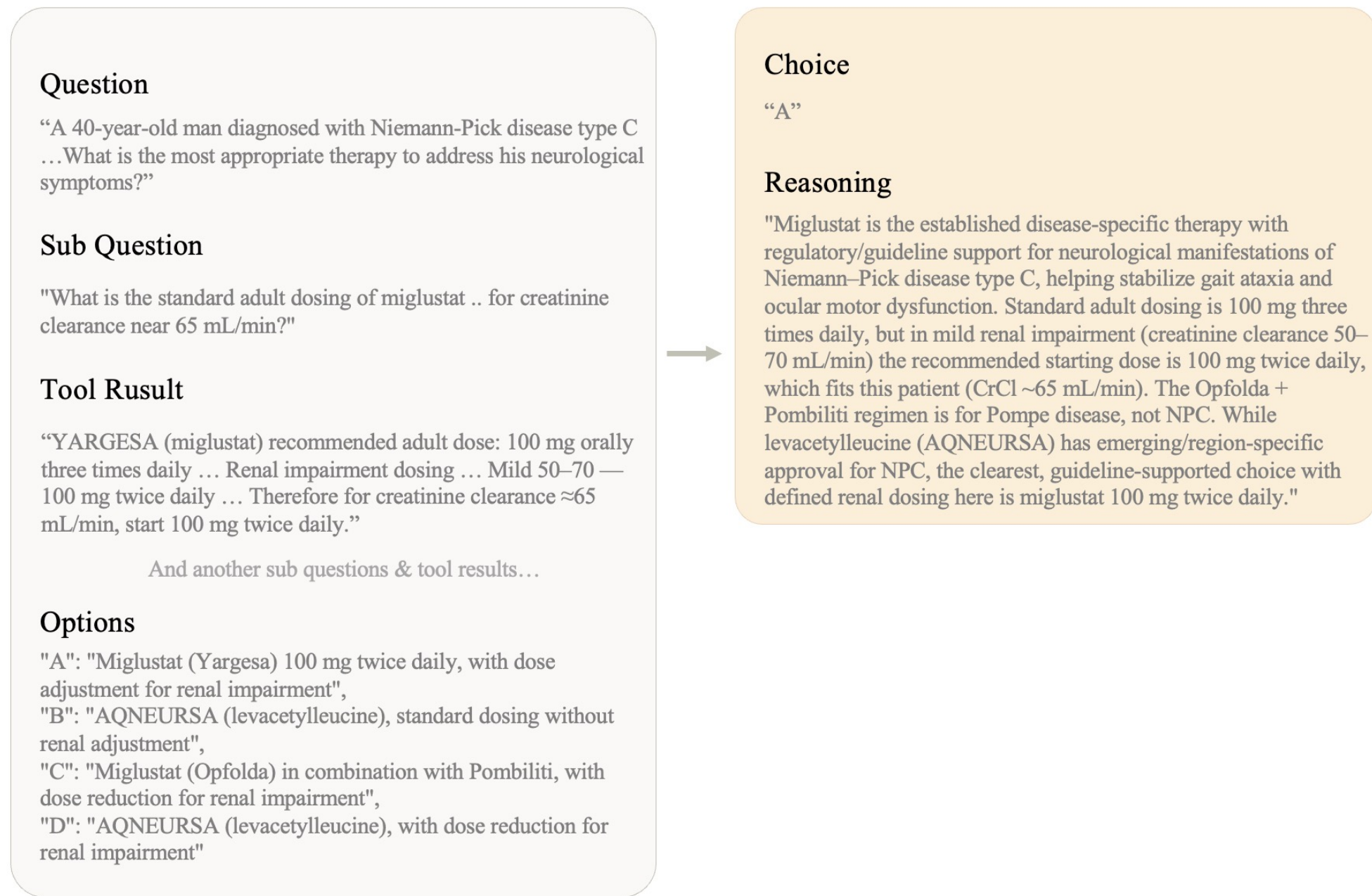
Break down the task into sub-questions using the given question and answer options.

Tool Utilization



Use each sub-question as a query to search for and invoke tools from the ToolUniverse via ToolRAG. Then verify whether each tool result is helpful for solving the main question using an LLM. If the verification returns false, repeat the process of searching for another tool, invoking it, and verifying the result again.

Answer Generation



Only the tool results that passed verification are recomposed to generate the final reasoning and answer.

4. Result

